Final Proposal: Effect of Continual Planning on Robotic Arm Catching Robustness

6.4210, Robotic Manipulation, Team 4.4

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1. Project Description

We will build a Drake simulation in which a single KUKA iiwa 14 robotic arm tosses a sphere and catches it with its gripper, a task that requires a lot of dexterity. Our core question is how the arm should use its observations of the ball to guide the catch and timing of the gripper closing, so we will compare a one-shot prediction with an approach that uses continual re-planning on identical tosses. Our project aims to demonstrate repeated toss-and-catch cycles, along with a quantitative comparison of catch success rate and consecutive successful cycles. We will measure the robustness of each approach by adding measurement noise and toss variability to simulate errors and latency that can arise in real environments.

2. Motivation

Tossing and catching an object is a highly dexterous dynamic manipulation: once the ball leaves the gripper, the arm has a short amount of time to reposition and intercept it, with no way to correct the ball's path. With that short amount of time, even small latencies in the throws and catches will impact whether the ball is caught or not. Tossing and catching are building blocks for passing objects, juggling, and catching falling items, with potential applications in home or industrial robotics. In general, tossing and catching push the limits of robot dexterity, and advancing the robustness of robotics in high-dexterity applications has far-reaching consequences across many applications of robotics. Prior work suggests that prediction quality often limits catching success [1]. Because the same arm both tosses and catches, any release error becomes the catcher's problem, which makes the comparison between a one-shot catch plan and a continually updated plan especially meaningful.

3. Related Work

Throwing has been addressed with physics-based optimization of joint trajectories [2], with reinforcement learning for underactuated machines [3], and with learned residuals on top of an analytic throw [4]. For catching, Lampariello et al. formulate real-time optimal catching trajectories [5], and Gold et al. combine trajectory prediction, catch-pose selection, and motion planning in one MPC problem, finding that success correlates with prediction accuracy [1]. Bombile and Billard show a dual-arm system grabbing an object while it is in motion [6], and one-arm juggling [7] is the closest precedent for a single arm repeatedly tossing and catching.

Our contribution is a controlled comparison of catching strategies under varying toss and sensing uncertainty, and a repeated-cycle evaluation. By combining variable tosses with sensing uncertainty, we hope to evaluate the robustness of one-shot vs continual motion planning in order to show when and how much continual planning improves catching accuracy. This approach follows from [1], but we also aim to test reliability under uncertainty similar to the uncertainty that robots will face in the real world due to small sensing inaccuracies.

4. Approach

Setup. One iiwa 14 arm with a gripper, and a sphere modeled as a rigid body. The toss is mostly upward but includes small lateral velocity variation, so the ball does not always come back to the hand. Ball position is observed with a small amount of simulated Gaussian noise, which is a standard method to approximate real-world sensor inaccuracy [8]. The catch intercepts the parabola where it crosses a catch plane at a chosen height, as in [1]'s use of intersection with a plane.

Toss. Choose a toss height that gives a flight time of about 0.5 to 0.65 seconds. Compute the release state from projectile motion, then generate a joint trajectory that reaches it under joint limits via nonlinear optimization [2].

Catch. The robot will start to close its gripper when the ball gets within an ideal distance to the gripper, which we will tune through simulation. Simulate latent delays that exist in the real world by adding small, fixed deltas from the time the robot decides to close the gripper to when it actually starts closing the gripper.

Approach A (predict once, then intercept). Fit a ballistic trajectory by least squares over an early window of noisy measurements, compute one intercept pose and time, and execute one planned motion. Provides a reference for the closed-loop replanning.

Approach B (closed-loop replanning). Refit the trajectory estimate at every control step (recursive least squares or a Kalman filter) and replan toward the updated intercept as data arrives, in the spirit of [1] but scaled down.

Reference baseline. An open-loop catcher that uses no observations and goes to where the nominal toss should land, to show how much each approach gains from sensing.

Evaluation. Evaluate the tradeoffs between the two approaches. Report catch success percentage out of 100 tosses, subject to different variables such as initial speed, toss height, and movement needed from ball release to catch. Also report success percentage of whether the gripper made contact with the ball.

Metrics will be evaluated across all three approaches (including the baseline), with different amounts of observation noise and toss variability, and results compared.

5. Applicable Topics from Class

- Kinematics and inverse kinematics (reachability from release pose to catch pose, joint limits)
- Trajectory generation and optimization (toss and intercept motions)
- Dynamics and contact (projectile flight, gripper closing on the ball)
- Grasping (grasp for a clean release, and tolerance of the catch grasp)
- State estimation and perception (trajectory fitting from noisy data)
- Control (tracking planned joint trajectories)

6. Project Goals and Timeline

Risks and scope. Joint velocity limits may cap the release speed and therefore the toss height, so we will check the limits for our model in week 1 and may relax them in simulation. We model the sphere without spin (and without drag).

Goals. Our baseline goal is to achieve at least 50% catch rate on the nominal catch with no noise in the toss and at least 90% contact rate with the gripper. We will also attempt to achieve successful trials with more noise in the toss. We anticipate observing a substantial increase in the accuracy of catching for the continual motion-planning approach compared to the one-shot approach.

Division of work. We aim to work on this project collaboratively, but we can think of a rough division of labor among our three group members as follows:

- **Member 1:** build environment, noise models, evaluation methods, and the trajectory estimator; later Approach B.
- **Member 2:** tossing (release state, trajectory optimization, release grasp).
- **Member 3:** Approach A catching (intercept planning, catch orientation, gripper timing), and the reference baseline.

Timeline

- **Week 1** (October 26):
    - project setup: GitHub, Drake environment with 1 arm and 1 sphere
    - Initial exploration into constraints, limitations, or assumptions we will make
    - All team members familiar with setup and pushing to github
    - Outline paper structure
- **Week 2** (November 2):
    - Work on Approach A: predict trajectory and execute planned motion
    - Roughly start or plan implementation for Approach B: closed-loop replanning
- **Week 3** (November 9):
    - Finish and refine working versions of our 2 approaches
    - Implement evaluation metrics and test with initial samples, finalize which metrics we plan to use
    - Add noise to both the toss and the observations to measure robustness of approaches
    - Project video storyboard
- **Week 4** (November 16):
    - Produce results: with 2 approaches working and our established metrics, run both approaches on ~100 tosses each and record results
    - Make any additional visualizations of results for paper
    - Start drafting final paper
- **Final weeks** (November 23):
    - Work on final paper
    - Record and edit final video

Final Project Deadline: Wednesday December 2

References

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